

On the Trade-offs of LLM-Augmented Multimodal Foundation Models for Clinical Image Classification

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Abstract

Multimodal foundation models enable joint visual and textual reasoning for clinical image interpretation, yet their behavior under large language model (LLM) augmentation remains insufficiently understood. We analyze the trade-offs introduced by augmenting vision-language models with LLM-generated clinical reports for chest X-ray classification. We adopt an inference-time augmentation strategy in which pseudo-reports generated by MedGemma are fused with image embeddings from BiomedCLIP. We evaluate image-only, LLM-augmented, and LLM-only configurations on the MIMIC-CXR-JPG dataset across 13 pathology labels. LLM augmentation improves specificity, Hamming accuracy, and exact-match accuracy, but reduces sensitivity, indicating a shift in decision behavior. Hallucination-aware metrics, including fabrication error rate and omission rate, quantify how errors in generated text propagate to classification outcomes. Results show reduced over-prediction but increased missed findings, especially for subtle pathologies. These findings highlight a reliability-dependent trade-off and motivate evaluation frameworks that jointly assess accuracy and clinical risk.

1. Introduction

Automated analysis of chest X-rays is a central problem in clinical AI. Advances in deep learning for radiology natural language processing [14, 18] and the rapid adoption of large language models (LLMs) [1] have transformed report generation and structured finding extraction. Multi-label classification remains challenging: each image may contain multiple pathologies, with class imbalance and frequent label co-occurrence. Subtle radiological features are also difficult to capture using visual representations alone. A recent survey by Liu et al. [12] identifies five generations of report generation methods, highlighting a shift from retrieval- and attention-based approaches toward LLM-driven pipelines that produce clinically coherent free-text findings.

Contrastive language-image pre-training (CLIP) has emerged as a powerful framework for joint visual-linguistic representation learning. Prior work shows that CLIP-based alignment enables zero-shot medical report generation without paired supervision [7]. BiomedCLIP [19] extends this approach to the biomedical domain using 15 million image-text pairs, learning embeddings that encode both visual and clinical semantics. Chen et al. [3] further show that improved visual-language alignment enhances diagnostic accuracy. When fine-tuned with an MLP classifier, BiomedCLIP produces strong representations for multi-label classification. However, in practice, ground-truth reports are unavailable at inference time, leaving the text encoder largely unused.

An underexplored approach is to provide generated text together with the image at inference time, allowing BiomedCLIP to use its full joint embedding space. Li et al. showed that instruction-tuned multimodal LLMs trained on biomedical image-text pairs can perform open-ended biomedical visual question answering, including chest X-ray interpretation [10]. Recent multimodal pipelines also show strong performance. Han et al. proposed the DFFG-MLLMR framework, which combines visual features with LLM-generated descriptions and achieves state-of-the-art results on MIMIC-CXR [5]. Liu et al. showed that CheXbert-derived pseudo-labels can guide supervised training, improving report generation by over 20% [13]. Dorfner et al. further showed that GPT-4 achieves high micro-F1 for zero-shot CheXpert label extraction [4]. These results support our hypothesis that MedGemma-generated pseudo-reports, encoded by BiomedCLIP, can enrich joint representations and improve multi-label classification.

Hallucination is a major barrier to using multimodal LLM-based systems in clinical settings. Artsi et al. reviewed 15 radiology studies and found that all exhibited hallucinations or misdiagnoses, reporting a fabrication rate of 1.47% and an omission rate of 3.45%, and showed that domain-specific prompting can reduce major errors by 75% [2]. To address this issue in chest X-ray report generation, Jiang et al. proposed chain-of-medical-thought

076	(CoMT) prompting, which decomposes diagnostic reason-	126
077	ing into structured steps [8]. These findings highlight a key	127
078	concern in our study: if pseudo-reports contain fabricated	128
079	or missing findings, these errors can directly affect down-	129
080	stream classification. We therefore introduce fabrication er-	
081	ror rate (FER) and omission rate (OR) to quantify how such	
082	errors propagate through the pipeline.	
083	This work presents a systematic three-way ablation	
084	study. We first use image-only BiomedCLIP embeddings	
085	with a trained MLP classifier. We then combine image em-	
086	beddings with pseudo-reports in an LLM-augmented set-	
087	ting fed into a scaled MLP. We also evaluate a zero-shot	
088	setting where MedGemma directly predicts labels without	
089	any trained classifier, following the LLM-only evaluation	
090	paradigm studied by Nowak et al. [15], who showed that	
091	open-weight LLMs can match GPT-4o in chest radiogra-	
092	phy finding extraction. In addition to standard classification	
093	metrics, we evaluate FER and OR to measure how halluci-	
094	nation affects classification decisions.	
095	Contributions. We develop a two-phase training pipeline	
096	for BiomedCLIP with an MLP classifier combined with	
097	inference-time LLM text augmentation. We conduct a com-	
098	prehensive ablation study on MIMIC-CXR-JPG revealing	
099	a clear trade-off between specificity and sensitivity under	
100	LLM augmentation. We provide per-label analysis across	
101	13 CheXpert pathology classes, highlighting patterns of im-	
102	provement and degradation. We introduce hallucination-	
103	aware metrics FER and OR, connecting pseudo-report qual-	
104	ity to classification performance.	
105	2. Related Work	
106	2.1. Medical Image Classification with CLIP	
107	CLIP-based models show strong zero-shot generalization	
108	in medical imaging. BiomedCLIP extends CLIP to the	
109	biomedical domain using 15 million image-text pairs and	
110	a dual-encoder design with a ViT-B/16 image encoder and	
111	a PubMedBERT text encoder. It supports both zero-shot	
112	inference and efficient fine-tuning. Prior work shows that	
113	fine-tuning the image encoder with a classification head	
114	achieves competitive performance on chest X-ray bench-	
115	marks, often outperforming convolutional baselines with	
116	fewer labeled samples [19].	
117	2.2. Multimodal LLMs in Radiology	
118	The emergence of general-purpose multimodal LLMs for	
119	medical imaging has expanded the capabilities of clinical	
120	AI. MedGemma [17], built on the Gemma 3 architecture	
121	and fine-tuned on diverse medical imaging data, can gen-	
122	erate free-text radiology descriptions in a zero-shot setting.	
123	Unlike earlier discriminative models, it enables open-ended	
124	clinical interpretation. Prior work on radiology report label-	
125	ing shows that GPT-4 and competitive open-source LLMs	
	achieve micro-F1 scores above 0.96 for extracting struc-	126
	tured labels from existing reports [4]. In contrast, our work	127
	uses MLLMs to generate pseudo-reports directly from im-	128
	ages, rather than relying on ground-truth text.	129
	2.3. Hallucination in Medical MLLMs	130
	Hallucination, defined as the generation of factually incor-	131
	rect or unsupported content, remains a major concern for	132
	LLMs in clinical settings. In medical imaging, it often ap-	133
	pears as fabrication (nonexistent findings are reported) or	134
	omission (true findings are missed). Prior work has exam-	135
	ined these behaviors in medical LLMs and emphasized the	136
	need for careful clinical validation before deployment [2].	137
	In this work, we define FER and OR as quantitative mea-	138
	sures of these failure modes, enabling direct assessment of	139
	how hallucination affects downstream classification perfor-	140
	mance.	141
	2.4. Text Augmentation for Vision Models	142
	Text augmentation improves visual representations by us-	143
	ing generated or retrieved text in semi-supervised learning	144
	and knowledge distillation. Vision-language models such	145
	as CLIP [16] learn joint image-text embeddings for cross-	146
	modal alignment. Recent work uses LLM-generated de-	147
	scriptions during training to enrich visual features, includ-	148
	ing frameworks such as LLaVA and LLaVA-Med [10, 11].	149
	In contrast, our approach focuses on inference-time aug-	150
	mentation, combining image and text embeddings directly	151
	in the CLIP space without additional fine-tuning.	152
	3. Methodology	153
	3.1. Dataset	154
	We use the MIMIC-CXR-JPG dataset [9] with labels de-	155
	rived from the CheXpert labeler [6], covering 13 pathology	156
	classes: atelectasis, cardiomegaly, consolidation, edema,	157
	enlarged cardiomeastinum, fracture, lung lesion, lung	158
	opacity, pleural effusion, pleural other, pneumonia, pneu-	159
	mothorax, and support devices. The test set contains 634	160
	studies, including 218 normal cases and 416 positive cases	161
	(at least one positive label).	162
	3.2. Pseudo-Report Generation	163
	Each PA chest X-ray is processed by MedGemma to gen-	164
	erate a free-text pseudo-report describing radiological find-	165
	ings. MedGemma is prompted with a standardized instruc-	166
	tion requesting a structured description of cardiopulmonary	167
	findings. No ground-truth reports from MIMIC-CXR are	168
	used; the pseudo-reports are entirely LLM-generated from	169
	the images. The same pseudo-reports are used consis-	170
	tently across all three ablation conditions. MedGemma out-	171
	puts serve as text inputs for the BiomedCLIP text encoder	172

(LLM-augmented condition) and as direct classification inputs (LLM-only condition, by parsing the generated text for label assertions).

3.3. Model Architecture

We use BiomedCLIP as the multimodal backbone and attach a lightweight MLP classifier for 13-label prediction. BiomedCLIP encodes the chest X-ray and the pseudo-report into 512-dimensional image and text embeddings, respectively. After L2 normalization, the two embeddings are concatenated into a 1024-dimensional fused feature. The classifier consists of two hidden layers (512 and 256 units), each followed by LayerNorm, GELU activation, and dropout. A final linear layer outputs 13 logits; sigmoid activation is applied for multi-label prediction with binary cross-entropy loss.

3.4. Two-Phase Training

Training is performed in two phases. In the first phase, the BiomedCLIP backbone is frozen and only the MLP classifier is trained for 30 epochs using Adam with a learning rate of 10^{-3} . In the second phase, the entire model is unfrozen and fine-tuned end-to-end for 20 epochs at 10^{-5} . Class imbalance is handled using weighted sampling throughout training.

3.5. Evaluation Metrics

We evaluate multi-label classification using macro and micro F_1 , sensitivity, specificity, Youden’s J, ROC-AUC, Hamming accuracy, and exact-match accuracy. All metrics use a fixed threshold of 0.5.

We also introduce hallucination-aware metrics:

$$\text{FER} = \frac{\text{FP}}{\text{FP} + \text{TP}}, \quad \text{OR} = \frac{\text{FN}}{\text{FN} + \text{TP}}. \quad (1)$$

The fabrication error rate (FER) measures false positives among predicted positives; the omission rate (OR) measures missed true positives. We further report FER_{an} restricted to abnormal cases.

For statistical analysis, we use the Wilcoxon signed-rank test for per-label comparisons ($n = 13$), the chi-squared test for aggregated metrics, and Fisher’s exact test for hallucination-related metrics, with Bonferroni correction for three pairwise comparisons.

4. Results and Discussion

4.1. Classification Performance

Table 1 summarizes classification performance across the three settings. The image-only model shows high sensitivity but lower specificity, indicating a tendency to over-predict positive labels, which is reflected by a high fabrication error rate.

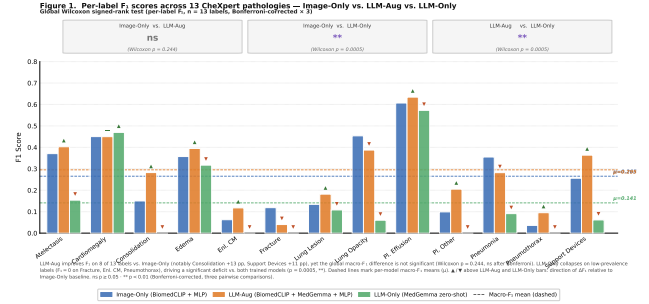


Figure 1. **Per-label F_1 comparison across three chest X-ray classification conditions.** Grouped bars show per-label F_1 on the MIMIC-CXR-JPG test set for Image-Only (BiomedCLIP + MLP), LLM-Aug (BiomedCLIP + MedGemma + MLP), and LLM-Only (MedGemma zero-shot). Dashed lines mark per-model macro- F_1 means (μ). ▲/▼ markers indicate the direction of ΔF_1 vs. the Image-Only baseline. Global Wilcoxon signed-rank test (two-sided, $n = 13$, Bonferroni-corrected): Image-Only vs. LLM-Aug $p = 0.244$ (ns); Image-Only vs. LLM-Only $p = 0.0005$ (**); LLM-Aug vs. LLM-Only $p = 0.0005$ (**).

The LLM-augmented model improves precision-related metrics, including micro F_1 , Hamming accuracy, and exact-match accuracy, while reducing fabrication errors. However, it significantly decreases sensitivity and increases omission, with no meaningful change in Youden’s J or ROC-AUC.

Overall, LLM augmentation shifts the model toward higher precision but lower recall, revealing a clear trade-off between over-prediction and missed findings.

4.2. Per-Label Analysis

Figure 1 shows per-label F_1 across the three settings. LLM augmentation improves most labels, while the LLM-only setting collapses on several classes. Gains are strongest for labels with clear visual and lexical cues. Consolidation improves through reduced false positives, while support devices and pleural findings benefit from consistent descriptions.

Performance degrades for subtle or broad labels. Fracture shows a sharp drop due to omission, while pneumonia and lung opacity decline due to semantic mismatch. Some labels show higher specificity but lower sensitivity, reflecting a precision-oriented shift. The LLM-only setting is biased toward negative predictions, leading to near-zero performance on several labels, while cardiomegaly remains stable. These trends align with FER and OR: reduced fabrication improves precision, but increased omission leads to missed findings.

4.3. Hallucination Metrics

FER and OR show a consistent pattern across the three settings. The image-only model exhibits high fabrication and

Table 1. **Classification performance on the MIMIC-CXR-JPG test set.** ROC-AUC is not reported for the LLM-Only setting due to the absence of probability outputs. Bold values indicate best performance per metric. Significance superscripts indicate statistical comparison vs. the Image-Only baseline (ns: $p \geq 0.05$; *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$; Bonferroni-corrected).

Metric	Image-Only	LLM-Aug	LLM-Only
Macro F ₁	0.265	0.295 ^{ns}	0.141 ^{**}
Micro F ₁	0.295	0.388 ^{***}	0.288 ^{***}
Hamming Acc.	0.620	0.818 ^{***}	0.892 ^{***}
Exact Match Acc.	0.156	0.257 ^{***}	0.284 ^{***}
Sensitivity (Macro)	0.735	0.456 ^{***}	0.127 ^{***}
Specificity (Macro)	0.603	0.839 ^{***}	0.967 ^{***}
Youden’s J	0.338	0.295 ^{ns}	0.094 ^{**}
ROC-AUC (Macro)	0.738	0.729 ^{ns}	—
ROC-AUC (Micro)	0.755	0.783 ^{ns}	—
FER	0.817	0.704 ^{***}	0.557 ^{***}
FER _{an}	0.773	0.647 ^{***}	0.441 ^{***}
OR	0.225	0.437 ^{***}	0.786 ^{***}

Wilcoxon signed-rank test (per-label, $n = 13$): Macro F₁, Sensitivity, Specificity, Youden’s J. Chi-squared test (aggregated): Micro F₁, Hamming Acc., Exact Match Acc. Fisher’s exact test: FER, FER_{an}, OR. ROC-AUC: Img vs. Aug Wilcoxon $p = 0.273$ (ns). FER_{an} = FER restricted to cases with ≥ 1 positive label; OR = omission rate = $1 - \text{Sensitivity}$.

low omission, reflecting a recall-oriented decision boundary. The LLM-only model reduces fabrication but misses most true findings, indicating a strong bias toward negative predictions. The LLM-augmented model lies between these extremes, reducing fabrication while increasing omission. This reflects a shift in the specificity–sensitivity trade-off rather than a simple threshold change, with similar ROC-AUC indicating preserved ranking performance.

A similar trend holds for abnormal-case FER, where LLM augmentation reduces over-prediction without the severe omission seen in the LLM-only setting. Overall, LLM augmentation improves precision but increases omission, revealing a trade-off between hallucination control and clinical sensitivity.

4.4. Analysis of the Specificity–Sensitivity Trade-off

The shift toward higher specificity and lower sensitivity under LLM augmentation can be explained by the embedding geometry of BiomedCLIP. The model aligns images and text in a shared space. When the pseudo-report accurately reflects the image, the fused embedding strengthens the positive signal and improves precision. When findings are under-described, the text embedding introduces a conflicting signal that suppresses positive predictions and reduces sensitivity.

This suggests that pseudo-report quality modulates the trade-off. Augmentation is most effective for visually salient and frequently described findings, where the text signal is reliable. For rare or subtle findings, omissions bias the model toward false negatives. From a clinical perspective, higher specificity benefits screening by reducing unneces-

sary follow-up, while higher sensitivity is critical for decision support. Future work may explore threshold calibration or label-specific strategies to balance this trade-off.

5. Conclusion

We study whether pseudo-reports generated by MedGemma can augment BiomedCLIP for chest X-ray multi-label classification. A three-way ablation reveals a clear specificity–sensitivity trade-off: LLM augmentation reduces false positives and improves agreement-based metrics, but increases false negatives, especially for rare pathologies. Hallucination-aware metrics, including FER and OR, link LLM output quality to classification behavior, showing that augmentation shifts the decision boundary without affecting ranking performance.

These findings have clinical implications. Higher specificity may benefit screening, while reduced sensitivity may limit decision support. This highlights the need to control hallucination and omission. More broadly, hallucination evaluation is essential for assessing multimodal models beyond standard metrics.

Future work will improve pseudo-report reliability through confidence-aware fusion and label-specific calibration. The proposed metrics provide a general framework for evaluating reliability in CLIP–MLLM pipelines and support safer clinical deployment.

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